Mapping and Reporting

October 23, 2024

SurvMeth632

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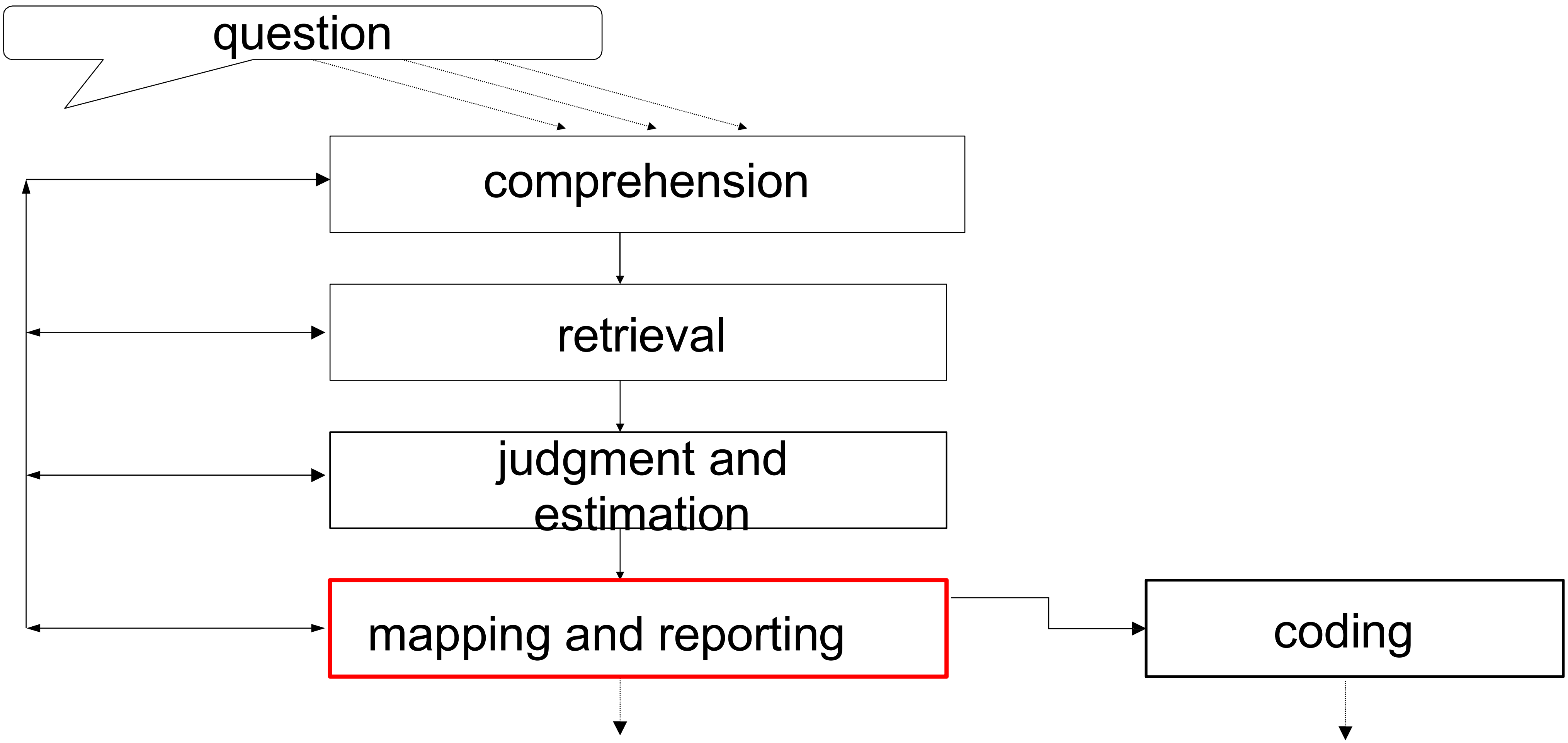
Mapping and reporting

- After *comprehension*, *retrieval*, and *estimation/judgment*, respondent must convert what is in mind into an answer
  - may not be expressed precisely
  - may need to be transformed into
    - a number
    - one (or more) of the options provided (if closed question)
  - may not match a response option
- Transformation to a response option can introduce error
- *Note: Will discuss open-ended verbal responses next week*


question
comprehension
retrieval
judgment and estimation
mapping and reporting
coding

Mapping and Reporting: Outline

1. Numerical (open) responses
   – Rounding, Visual Analog Scales
2. Ordered response scales
   – Extracting pragmatic information from scales
   – Conceptual vs. visual midpoint
3. Unordered/non-dimensional response scales
   – Serial position/response order effects
      • cognitive sophistication
      • working memory

Outline (cont'd)

4. Questionnaire features
   – Middle alternatives
   – "Don't know"

5. Editing responses: questions about sensitive/threatening topics
   – Deception?

Three kinds of response scales

1. Open numerical (implicit scale)
2. Ordered (dimensional)

   e.g., very satisfied … very dissatisfied
3. Unordered (non-dimensional) – list of options

   e.g., Amazon Music, Tidal, Spotify, Apple Music, YouTube Music, …

- In (1) and (2), *R*s likely to mentally represent scale; reduces need to memorize options (if read by interviewer) because immediately evident once response dimension is known
- In (3) categories must be remembered or visually persistent because cannot be derived from position on a dimension

Open Numerical Responses: Rounding

- When *R* has only a qualitative sense of magnitude (e.g., “a lot”), i.e., no actual numerical info, mapping to a numerical value may introduce error
  - “Last year, how many times did you...?”
- One example is *prototypical* or *rounded* reports
  - e.g., in estimates of elapsed time or very frequent behaviors
  - may indicate imprecision in underlying representation
  - may simplify mapping task by creating “categories” of numbers or ranges
  - may signal uncertainty or embarrassment
- Rounded responses generally assumed to reflect estimation rather than recall-based reporting
  - more likely to involve measurement error than unrounded numbers

Rounding (2)

Evaluation of candidates on 100 point feeling thermometer

	Number responses	% total
Clinton		
Multiples of 10	1907	78.9
15, 85	310	12.8
All other values	199	8.2
Bush		
Multiples of 10	1948	79.3
15, 85	412	16.8
All other values	98	4.0

American National Election Studies, 1992;
source: Tourangeau, Rips & Rasinski, 2000

Rounding (3)

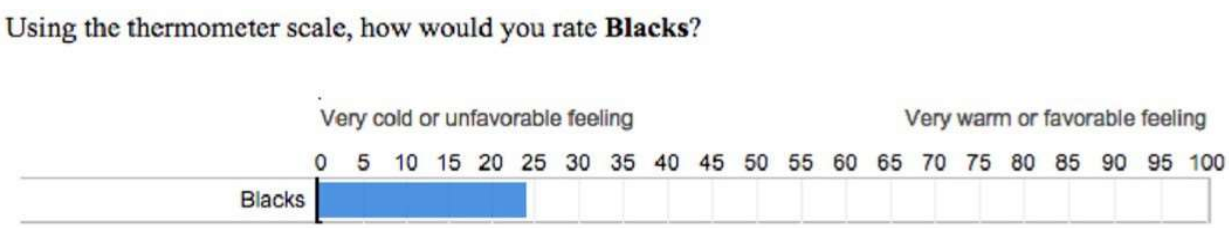
- May signal reliance on non-numerical (and thus imprecise) information
  - When only a qualitative impression of frequency (e.g., “pretty often”) is available, rounded responses more frequent than when *R*s enumerate or use rates (Conrad, Brown & Cashman, 1998)
- May signal embarrassment
  - Tourangeau & Smith (1997,1998) asked *R*s about number lifetime sex partners
  - most peaks in response distribution at multiples of 5
  - Rounding especially likely when number of partners high: 64% of responses that were greater than 10 were multiples of 5
    - rounded reports may be *intended to convey* “fuzziness” of response or use of a range, even if actual numerical values known
    - may conceal actual, embarrassing value; allows *R* to hide behind seemingly imprecise response

Rounding (4)

- But could be sheer magnitude
  - Difficult to recall and count 50+ partners
  - Not about sensitivity of response per se
- Consequences for accuracy
  - Reduces precision
  - Downward bias, even if round to nearest prototypical value
    - intervals around prototypical values are asymmetrical so that more values are rounded down than up (Huttenlocher, Hedges & Bradburn, 1990):
    - e.g., 10, 25, 50, 100, 500, 1000, 10000 …
    - So, reports of 50 may include actual values rounded down from 75 and rounded up from 37.5, i.e., more values rounded down than up

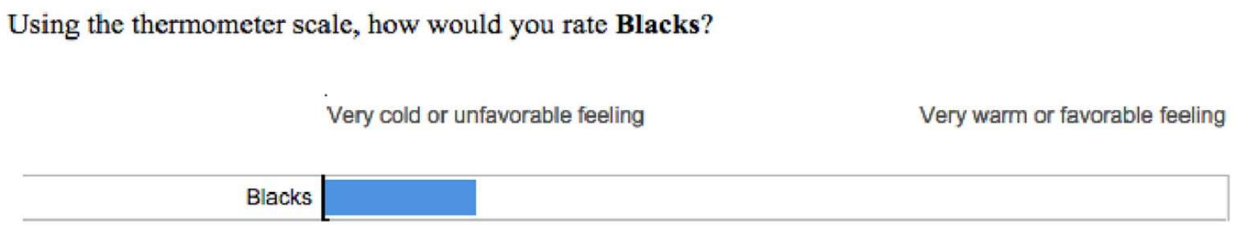
Rounding and Visual Analog Scales

- Visual analog scales (VAS) represent entire number line, usually from 0 to 100, i.e., 101-point scale
  - allow *R*s to pinpoint their response
  - One problem: *R*s may not know exactly what value they have selected

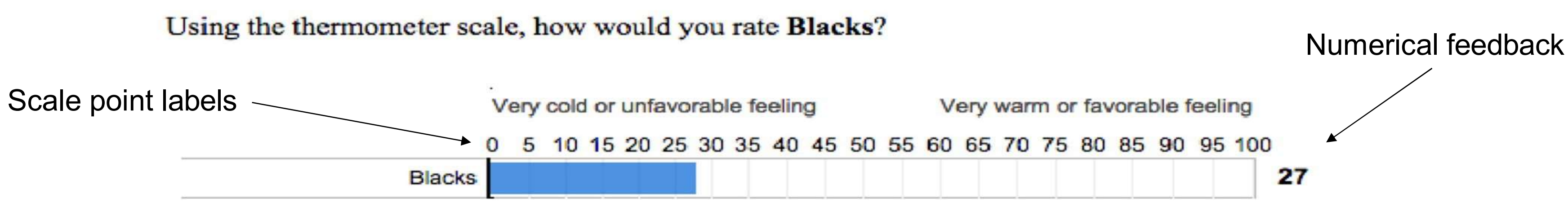


Rounding and VAS (2)

- Liu & Conrad (2016) measured rounding with 4 designs: 4 VASs, a dropdown menu, and a text box
  - VAS either displayed no feedback about selected value (top) or displayed feedback (bottom), and scale points were either not labeled (top) or were labeled in intervals of five (bottom)



Experimental condition 2. VAS: No Numerical Feedback, End Labels Only.



Experimental condition 3. VAS: Numerical Feedback, Labeled at Regular Intervals.

Rounding and VAS (3)

- The more explicit the numerical info, the more rounding

Feeling Thermometer Group						
	No Numeric Label		Numeric Label		Drop Down	Text Box
Feedback	Yes	No	Yes	No	--	--
Blacks	52	43	67	62	76	95
Whites	57	45	72	69	76	94
Hispanics	58	44	66	60	75	95

% rounded responses

Liu & Conrad (2016)

- Suggest *R*s may want to round but VAS interface makes difficult

Ordered Response Scales: Symmetrical Attitude Scales

- Most widely used procedure for measuring attitudes is “Likert scale”
- *R*s asked to place themselves on a scale, one end of which is positive (e.g., favor, agree) and the other negative (e.g., oppose, disagree)

Ordered Response Scales: Symmetrical Attitude Scales (2)

The United States spends too much money on scientific research

Please tell me how much you disagree or agree with this statement by choosing a number between 1 and 7,
where 1 means "strongly disagree" and 7 means "strongly agree."

Agreement Scales and Response Style

Response Style: selecting positive or extreme or middle values irrespective of what it means – considered measurement error

- *Acquiescence bias*: tendency to agree or respond "yes"
  - May smooth the interaction because relatively non-controversial; to the extent this short circuits the response process, considered a type of *survey satisficing**
- *Extreme response style*: tendency to select scale endpoints, possibly to convey authority
- *Middle response bias:* tendency to select middle category, possibly to convey neutral or non-committal stance

- More common in some ethnic, cultural, and racial groups than others – may introduce differential measurement error

*topic of week 11 class

Ordered response scales: Numerical Labels

- Numerical scale values, especially end points (anchors), can affect interpretation of verbal labels (Schwarz et al. 1991)

  "How successful would you say you have been in life?"

  |- - -|- - -|- - -|- - -|- - -|- - -|- - -|- - -|- - -|- - -|

  not at all successful — extremely successful

  -5 — +5

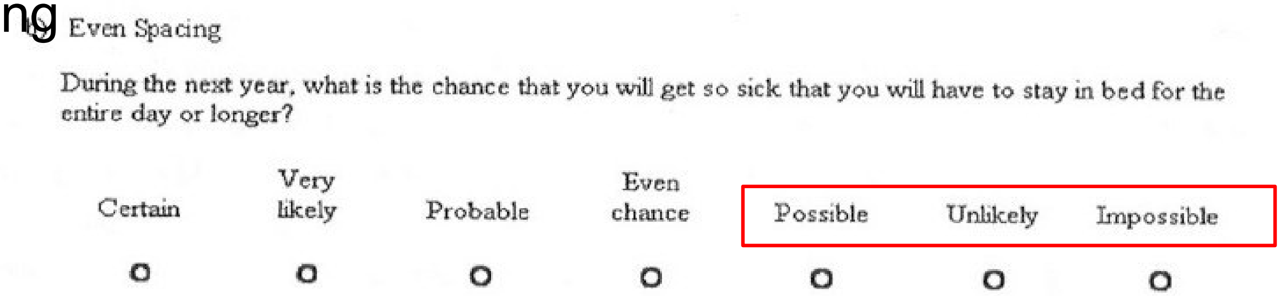
  – 34% of 0 to 10 group responded in lower half (0 to 5)
  – 13% of -5 to +5 group responded in lower half (-5 to 0)
  – zero scale anchor combined with label to mean "absence of success"
  – negative scale values combined with label to mean "presence of failure"

Ordered response scales: Conceptual vs. Visual Midpoint

- *R*s' mental representation of visually presented scale involves underlying concept <u>and</u> visual representation
- Tourangeau et al. (2004) compared endorsement of middle option when scale spaced evenly vs. unevenly
  - Uneven spacing made scale points on right side, conceptually, look more central

Conceptual vs. Visual Midpoint

Even spacing

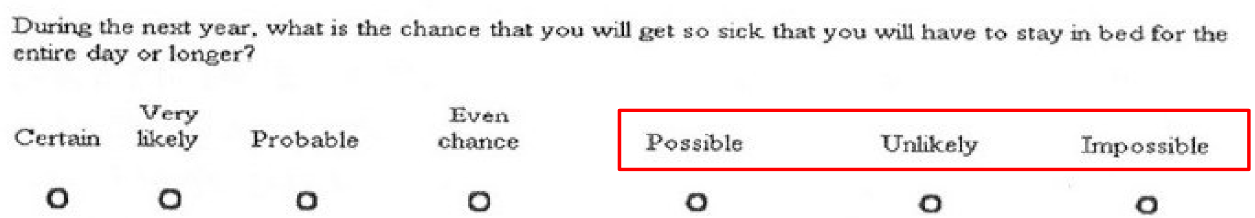

b) Even Spacing

During the next year, what is the chance that you will get so sick that you will have to stay in bed for the entire day or longer?

Certain	Very likely	Probable	Even chance	Possible	Unlikely	Impossible
O	O	O	O	O	O	O

58% *R*s selected option on (conceptual*) right side

Uneven spacing


During the next year, what is the chance that you will get so sick that you will have to stay in bed for the entire day or longer?

Certain	Very likely	Probable	Even chance	Possible	Unlikely	Impossible
O	O	O	O	O	O	O

64% *R*s selected option on (conceptual*) right side

*also right side visually

*visually covers more of scale than with even spacing

Ordered Response Scales: Non-differentiation or "straightlining"

- In battery of statement that use the same response scale for each, some *R*s select same response category for all or most questions
  - e.g., selecting "somewhat agree" for 7 out of 8 questions, even when some "reverse worded"

    Q71a. I enjoy work that requires the use of words.

    Q71c. I do a lot of reading.

    Q71g. I prefer activities that don't require a lot of reading.
  - Almost surely reflects reduced effort and, thus measurement error

Straightlining in a grid (matrix)

For each of the following statements, please rate how much it characterizes you.

	Extremely uncharacteristic	Somewhat uncharacteristic	Uncertain	Somewhat characteristic	Extremely characteristic
I prefer to avoid taking extreme positions	○	○	○	◉	○
I want to know exactly what is good and bad about everything	○	○	○	◉	○
If something does not affect me, I do not usually determine if it is good or bad	○	○	○	◉	○
There are many things for which I do not have a preference	○	○	○	◉	○
I like to have strong opinions even when I am not personally involved	○	○	○	◉	○
I would rather have a strong opinion than no opinion at all	○	○	○	◉	○
I only form strong opinions when I have to	○	○	○	◉	○
I am pretty much indifferent to many important issues	○	○	○	◉	○

- Can occur when battery presented in a grid (as above) or as individual questions, either by interviewer or online
- Straightlining more common among *Rs* with less education (e.g., Rogers and Herzog, 1984), and at end of questionnaire (Knowles, et al., 1981)

Unordered response scales: Response Order Effects

- When response options have no ordered relation, i.e., not arrayed on dimension, order of presentation can affect choices
- Mode (visual or verbal) affects preference for early or late items
  - lists of textual, i.e., visually presented, options lead to *primacy* effects, i.e., preference for options processed earlier
  - spoken lists of options lead to *recency* effects, i.e., preference for more recently processed options
- Response order effects sometimes observed for ordered scales: when opinions represented as ranges rather than scale points, first point within “acceptable” range endorsed (Krosnick & Presser, 2010)

Mode can influence response order effects

- Krosnick & Alwin (1987) argue that
  - under visual presentation *R*'s mind becomes "cluttered" after first few options
    - first few options – which are probably more clearly understood – are more likely to be endorsed
    - *primacy effect*
  - under spoken presentation mental representation of earlier options likely to be overwritten by later ones, or just forgotten (working memory capacity exceeded)
    - last few options more likely to be endorsed
    - *recency* effect

Krosnick & Alwin (1987)

- Visual (show card) presentation in FTF interviews
- Choose most important child characteristics from list of 13
- Two orders helps assure that bias toward early responses not due to their content

Quality	Standard order	Reversed order	Difference
	%	%	%
Manners	26.4	10.1	16.3
Success	19.1	14.6	4.5
Honest	65.7	48.4	17.3
.	.	.	.
.	.	.	.
Considerate	24.9	39.5	-14.6
Interested	17.9	24.9	-7.0
Studious	6.5	16.4	-9.9

Primacy and "Cognitive Sophistication"

- K&A found that primacy effect was only present for *R*s with "low sophistication" -- lower levels of education and vocabulary score
- argued that this group most likely to respond as soon as acceptable option encountered, i.e., to satisfice
  - note: "satisficing" is adapted from Herbert Simon's influential idea; applied broadly across problem solving situations and not survey responding
- Malhotra (2008) observed primacy effects most often for
  - *R*s low in education
  - who responded most quickly

Response Order Effects and Working memory

- Knäuper (1998, 1999) controlled for working memory capacity using respondent age as proxy
  - Reanalyzed Shuman & Presser (1981) studies on divorce and housing
    - older *R*s showed response order effects (primacy for divorce, recency for housing)
    - younger respondents rarely produced order effect
  - Ruled out education as account for findings
    - Because older adults generally have less education than younger adults, finding could potentially be due to education and not age
    - But Knäuper found (in 14 studies) age significantly predicted recency for both low and high education *R*s
    - Whereas education marginally significant for young *R*s and not sig. for older *R*s

Middle Alternatives

Should divorce in this country be easier or more difficult to obtain than it is now?		Should divorce in this country be easier to obtain, more difficult to obtain, or stay as it is now?	
Easier	28.9	Easier	22.7
More difficult	44.5	More difficult	32.7
Stay as is (volunteered)	21.7	Stay as is	40.2
Don't know	4.9	Don't know	4.3
Total	100%	Total	100%

- Whether or not middle alternative ("Stay as is") is explicitly offered, ratio of "Easier" to "More difficult" is about 2/3 (Shuman & Presser, 1981)

Don’t Know (DK) and No Opinion

- Researchers screen out non-opinions by offering DK or No Opinion response options
- Schuman & Presser (1981) and others have observed more DK responses when explicitly offered than when *R*s volunteer they don’t know
- *R*s more likely to report “no opinion” in response to explicit filter (“Do you have an opinion on X?)
- Increases with strength of filter (“Have you thought enough about X to have an opinion?”)
- Hippler & Schwarz (1989) show that strong filters interpreted as demand for extensive knowledge
  - may lead *R*s to falsely indicate they have no opinion to avoid more questions and increasingly difficult questions on the topic

Editing responses (social desirability)

- Reporting may be affected by topic of question
  - Behaviors viewed as positive, e.g., voting, generally overreported
  - Behaviors viewed as negative, e.g., drug use, generally underreported
- *R*s seem to edit what they report to avoid embarrassment, legal action, and other harm
- Sensitivity of topic can differently affect reporting depending on how it is perceived
  - Males (in some studies) report more female sexual partners than women report male partners (though should be the same if sampling is random)
    - Usual explanation: <u>more</u> partners is socially desirable for men, <u>fewer</u> partners is socially desirable for women

Editing responses: Self-administration

- Self administration almost always increases undesirable – presumably honest – reporting
  - e.g., Tourangeau & Smith (1997,1998) reduced male-female asymmetry in reports of different-sex partners with paper and computerized self administration
- Will discuss self-administration and disclosing sensitive information in more detail in week 10

Editing responses: Awareness

- Schaeffer (2000) points out that *R*s may not be aware of editing processes
  - For topics that are *no one's business*, people may report about "public persona" rather than refusing to answer
    - Is this deception?
  - for these cases, mode of administration should not matter
- Some topics are taboo regardless of a how one would respond

  e.g., "How much do you earn annually?"
- For others, threat depends on truth of one's circumstances, i.e., some answers are more threatening than others

  e.g., "Do you currently inject opioids?"
    - threat may depend on perceived likelihood of disclosure

Editing Responses: Motivated Misreporting

- Motivated misreporting (Tourangeau & Yan, 2008)
  - Evidence suggests that socially desirable reporting <u>is</u> the result of deliberate misreporting:
    1. Misreporting is in one direction
    2. Self-administration affects answers to sensitive but not non-sensitive questions
  - More likely result of editing a formulated response than selectively retrieving positive attributes about self
    - Holtgraves (2004) presented either socially desirable or undesirable self-descriptions (adjectives) and timed participants as decided if descriptions applied to themselves or not
    - response times longer for undesirable than desirable descriptions
  - Suggests *R*s consider editing answers, increasing RT

Summary

- At least 2 main sources measurement error in response selection
  1. Characteristics of the response options and response scales
     a) Open numerical responses are often rounded
     b) Ordered options (scales): how *R*s mentally represent dimension affected by numerical labels and spacing/symmetry
     c) Ordered scale for battery of items can produce straightlining
     d) Unordered options can produce response order effects
        – primacy with visual presentation, recency with spoken presentation
     e) Middle options and DK options selected when provided but may “draw” equally from both sides of scale

Summary (2)

2. Socially desirable responding
   a) Self-administration promotes candid responding
   b) Some items inherently sensitive (e.g., income, religion); others depend on true value (e.g., drug use)
   c) Instead of deception, may be “good faith” response about public persona
   d) But evidence suggests misreporting is deliberate and is due to editing responses